

MHopReg: Efficient Hierarchical Multi-Hop Graph Search for Point Cloud Registration

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Abstract

Outlier rejection for correspondence-based point cloud registration confronts two fundamental challenges in real-world scenarios. First, low-overlap regions yield sparse and fragmented inlier distributions that are difficult to discover using conventional one-step global search strategies. Second, large-scale scenes present dense correspondence inputs that impose stringent requirements on the accuracy-efficiency trade-off of search algorithms. To this end, we propose a hierarchical multi-hop graph search framework that progressively refines correspondences to address these challenges. Our method constructs a compatibility graph with transformation-invariant embeddings to predict correspondence confidence, establishing the foundation for cluster-balanced seed sampling that ensures comprehensive coverage across fragmented regions. These strategically selected seeds subsequently drive hierarchical multi-hop expansion, progressively discovering inliers through multi-resolution graph layers while circumventing the high complexity of exhaustive global search. Finally, distribution-aware ranking jointly evaluates geometric consistency and spatial coverage to select well-distributed transformations from multiple hypotheses. Experiments on 3DMatch, 3DLoMatch, and KITTI demonstrate that our method achieves highly competitive performance compared to existing methods in both low-overlap and large-scale scenarios.

1. Introduction

Point cloud registration is a fundamental problem in robotics perception, SLAM, autonomous driving, and 3D reconstruction [16, 30, 35, 42, 48]. Given two partially overlapping point clouds at different 3D positions, point cloud registration aims to estimate the relative rigid transformation for alignment [33]. With the advancement of

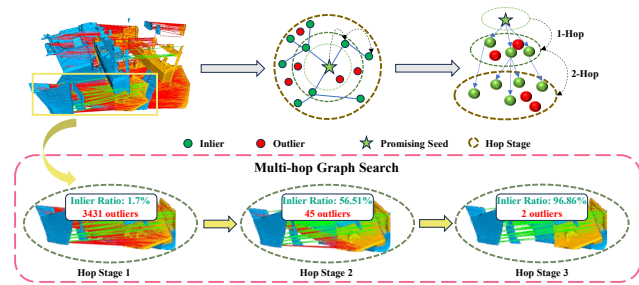


Figure 1. **Multi-hop graph search progressively refines correspondences.** Three-stage hierarchical expansion reduces outliers from 3431 to 2 (**99.94%**) while improving inlier ratio from 1.7% to 96.86% (**57x**).

deep learning and 3D point cloud understanding techniques, feature-based point cloud registration methods [3, 17, 22, 27, 34, 38, 41, 43] have developed rapidly and achieved remarkable results. However, in scenarios with low overlap, high symmetry, or large-scale outdoor environments, these methods inevitably generate a large number of incorrect correspondences [47], making the rejection of outlier correspondences a critical step for successful registration [4].

Existing mainstream outlier correspondence rejection methods mainly exhibit two major trends: geometry-based methods and deep learning-based methods [19, 47]. Geometry-based methods can be further divided into two categories: one involves iteratively sampling minimal subsets of correspondences [5, 12], while the other employs graph-based structures to search for locally consistent inlier sets based on spatial consistency [7, 8, 45]. In scenarios with low overlap and high symmetry, outliers tend to exhibit more local geometric consistency, causing geometry-based validation methods to fail. Although MAC [45] demonstrates the potential to capture fine-grained structures, this method is constrained by NP-hard search complexity [24, 46], which renders it inefficient for large-scale correspondence inputs. Deep learning-based methods [4, 18] for-

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mulate correspondence filtering as a classification problem and perform non-local inlier search in the global feature space using top-K high-confidence seed correspondences. Compared to PointDSC [4], although VBReg [18] fuses correspondence confidence through multi-layer voting, it still follows the traditional global ranking search framework, making it inherently difficult for initial seed correspondences to effectively retrieve highly scattered locally consistent correspondence subsets.

In this paper, our goal is to design a novel outlier correspondence rejection method to address the aforementioned challenges, which can handle the difficulties of searching in low-overlap scenarios while avoiding the high complexity of one-step global search with large-scale correspondence inputs. As shown in Fig. 1, the proposed method removes a large number of outliers in extreme outlier scenarios and substantially improves efficiency compared to global search by progressively filtering correspondences in a hierarchical manner.

Our method, named MHopReg, follows a hierarchical expansion framework composed of four main modules: $SE(3)$ -Equivariant Graph Encoding, Cluster-Balanced Seed Sampling, Hierarchical Multi-Hop Expansion, and Distribution-Aware Hypothesis Ranking. The $SE(3)$ -Equivariant Graph Encoding constructs a compatibility graph and learns transformation-invariant embeddings to predict correspondence confidence scores, which provide the foundation for subsequent seed selection. Building upon these learned confidences, the Cluster-Balanced Seed Sampling strategically selects spatially diverse seeds across different graph clusters to ensure comprehensive coverage in fragmented low-overlap scenarios. These selected seeds then serve as starting points for the Hierarchical Multi-Hop Expansion, which progressively discovers inliers through multi-resolution layers based on graph topology, avoiding the high complexity of one-step global search while ensuring structural consistency. Finally, the Distribution-Aware Hypothesis Ranking evaluates multiple hypotheses generated from different seeds by jointly considering inlier quantity, spatial distribution, and alignment precision to select the optimal transformation. Our main contributions are:

- We propose MHopReg, a hierarchical multi-hop graph expansion framework that efficiently handles high outlier ratios and large-scale scenes through progressive seed-frontier expansion.
- We introduce cluster-balanced seed sampling for diverse cluster coverage, hierarchical inlier propagation for progressive multi-resolution expansion, and distribution-aware hypothesis ranking for balancing inlier distribution and alignment precision.
- We achieve highly competitive performance on 3DMatch, 3DLoMatch and KITTI, attaining the highest registration recall on 3DLoMatch.

2. Related Work

2.1. Learning-based Feature Matching

Feature matching establishes the correspondence set for registration [1, 27, 40, 44]. Traditional hand-crafted descriptors such as FPFH [28] and SHOT [32] rely on manually designed geometric signatures, which are sensitive to noise and partial overlap. Deep learning has enabled more discriminative features: FCGF [9] employs fully convolutional networks for dense feature extraction, Predator [17] introduces overlap-attention modules to enhance robustness in low-overlap scenarios, GeoTransformer [27] leverages geometric transformers for super-point matching, and PareNet [38] incorporates rotation-equivariant encoding for improved discrimination. Despite these advances, feature matchers still produce 70–90% outliers in challenging scenes [17, 39], making robust outlier rejection a critical post-processing step.

2.2. Geometry-based Outlier Rejection

Classical methods optimize geometric alignment through local or global constraints. ICP [6] and its variants [29] iteratively minimize point-to-point or point-to-plane distances but require good initialization and are prone to local minima. RANSAC [12] and its successors [5, 11] employ hypothesis sampling and verification but suffer from exponential complexity when outlier ratios exceed 90%. Recent methods pursue global solutions through different paradigms: TEASER++ [36] achieves certifiable optimality via truncated least squares at $O(N^3)$ complexity, SC^2 -PCR [7] performs spectral ranking using second-order spatial consistency, and MAC [45] or FastMAC [46] formulate outlier rejection as maximal clique enumeration with progressive pruning. Alternatively, PCRegen [47] addresses the problem from a correspondence regeneration perspective, progressively reconstructing high-quality matches from initial noisy correspondences. Despite their effectiveness, these methods struggle to balance structural coherence with computational efficiency in high-outlier scenarios.

2.3. Learning-based Outlier Rejection

Deep learning methods model outlier rejection as confidence estimation or graph reasoning. DGR [10] formulates this as binary classification via deep graph neural networks. PointDSC [4] embeds spatial consistency into GNN-based feature aggregation and applies neural spectral matching for inlier confidence estimation, but under high outlier ratios ($>90\%$), spatial consistency signals become ambiguous, degrading non-local aggregation. Hunter [37] introduces hypergraph-based higher-order consistency for severe outlier scenarios, but its $O(N^3)$ hyperedge enumeration limits scalability. VBReg [18] leverages variational Bayesian in-

ference for probabilistic inlier evaluation. Despite their effectiveness, existing methods face a fundamental trade-off: geometry-based global spectral approaches achieve structural coherence but scale poorly, while learning-based local methods enable efficient search but struggle with fragmented inlier distributions. Our work addresses this via hierarchical multi-resolution expansion progressively building inlier hypotheses through structured graph exploration.

3. Method

3.1. Problem Formulation

In pairwise 3D registration, given a source point cloud $\mathcal{X} = \{\mathbf{x}_i \in \mathbb{R}^3 \mid i = 1, \dots, |\mathcal{X}|\}$ and a target point cloud $\mathcal{Y} = \{\mathbf{y}_j \in \mathbb{R}^3 \mid j = 1, \dots, |\mathcal{Y}|\}$, we aim to estimate the optimal rigid transformation $\mathcal{T}^* = (\mathbf{R}^*, \mathbf{t}^*)$ consisting of a rotation matrix $\mathbf{R}^* \in SO(3)$ and a translation vector $\mathbf{t}^* \in \mathbb{R}^3$ that aligns their overlapping regions. We construct putative correspondences $\mathcal{C} = \{c_k = (\mathbf{x}_k, \mathbf{y}_k) \in \mathbb{R}^6 \mid k = 1, \dots, N\}$ via feature-level nearest neighbor matching, where $N = |\mathcal{C}|$ denotes the total number of correspondences, and $\mathbf{x}_k \in \mathcal{X}$ and $\mathbf{y}_k \in \mathcal{Y}$ are matched keypoints. A correspondence is classified as an inlier if it satisfies the distance criterion $\|\mathbf{R}^* \mathbf{x}_k + \mathbf{t}^* - \mathbf{y}_k\|_2 < \tau_d$, where τ_d defines the inlier distance threshold. The primary challenge arises in low-overlap scenarios where the overlap ratio drops below 30% and the inlier rate falls under 10% [17, 45], causing traditional RANSAC-based methods to fail. The pipeline of our proposed method is shown in Fig. 2.

3.2. SE(3)-Equivariant Graph Encoding

Traditional correspondence feature learning methods [10, 26] apply generic operators that ignore the unique properties of 3D rigid transformations. We propose to explicitly encode the intrinsic geometric transformations between correspondences through an equivariant graph representation, where each correspondence is treated as a node and edges encode their spatial compatibility.

Compatibility Graph Construction. We construct the compatibility graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where nodes $\mathcal{V}$ represent correspondences in $\mathcal{C}$, by connecting pairs that exhibit geometric consistency under rigid transformations. Following prior work [4, 7, 45], we measure the distance compatibility $\Delta d_{ij} = \|\|\mathbf{x}_i - \mathbf{x}_j\|_2 - \|\mathbf{y}_i - \mathbf{y}_j\|_2\|$ as the absolute difference between inter-point distances in source and target clouds, and the angle compatibility $\Delta \theta_{ij} = |\arccos(\mathbf{n}_{\mathbf{x}_i} \cdot \mathbf{n}_{\mathbf{x}_j}) - \arccos(\mathbf{n}_{\mathbf{y}_i} \cdot \mathbf{n}_{\mathbf{y}_j})|$ as the difference between normal angles, where $\mathbf{n}_{\mathbf{x}_i}$ and $\mathbf{n}_{\mathbf{y}_i}$ denote surface normals at $\mathbf{x}_i$ and $\mathbf{y}_i$. An edge $(c_i, c_j) \in \mathcal{E}$ exists if both constraints are satisfied:

$$\mathcal{E} = \{(c_i, c_j) : \Delta d_{ij} < \epsilon_d \wedge \Delta \theta_{ij} < \epsilon_\theta\}, \quad (1)$$

where ϵ_d and ϵ_θ are distance and angle thresholds respectively. The graph distance $d_{\mathcal{G}}(c_i, c_j)$ is defined as the minimum number of hops between nodes c_i and c_j in $\mathcal{G}$.

SE(3)-Equivariant Message Passing. To extract transformation-invariant features while preserving the geometric structure, we employ spherical harmonic encoding [13, 23, 31]. A naive initialization using absolute coordinates $[\mathbf{x}_i; \mathbf{y}_i]$ lacks translation invariance. To ensure $SE(3)$ equivariance, we first compute the translation-normalized relative coordinates by subtracting the respective centroids of the input point clouds: $\Delta \mathbf{x}_i = \mathbf{x}_i - \bar{\mathbf{x}}$ and $\Delta \mathbf{y}_i = \mathbf{y}_i - \bar{\mathbf{y}}$. Each correspondence c_i is then initialized with a feature vector $\mathbf{h}_i^{(0)} = \text{MLP}([\Delta \mathbf{x}_i; \Delta \mathbf{y}_i])$.

While this normalization guarantees translation invariance, to further ensure strict rotation equivariance, the message passing leverages relative direction encoding. For each edge $(c_i, c_j) \in \mathcal{E}$, we compute the relative position vector $\mathbf{r}_{ij} = \mathbf{x}_j - \mathbf{x}_i$ in the source cloud and encode its directional geometry via spherical harmonics $\mathcal{Y}(\mathbf{r}_{ij})$. The equivariant message passing at layer ℓ is formulated as:

$$\mathbf{h}_i^{(\ell+1)} = \mathbf{h}_i^{(\ell)} + \sum_{j:(c_i, c_j) \in \mathcal{E}} \text{MLP}([\mathbf{h}_j^{(\ell)}; \mathcal{Y}(\mathbf{r}_{ij})]). \quad (2)$$

By explicitly decoupling translation invariance (via normalized coordinates) and rotation equivariance (via spherical harmonics), the learned embeddings $\tilde{\mathbf{h}}_i^{(L)} = \mathbf{h}_i^{(L)}$ strictly preserve the $SE(3)$ geometric properties crucial for robust outlier rejection.

3.3. Cluster-Balanced Seed Sampling

Using the learned $SE(3)$ -equivariant features $\{\tilde{\mathbf{h}}_i^{(L)}\}_{i=1}^{|\mathcal{C}|}$, we first predict the inlier confidence $\hat{s}_i \in [0, 1]$ for each correspondence c_i :

$$\hat{s}_i = \sigma(\text{MLP}(\tilde{\mathbf{h}}_i^{(L)})), \quad (3)$$

where σ is the sigmoid activation. Existing methods [4] typically apply spatial non-maximum suppression (NMS) [25] to disperse seeds physically. We adopt this to form a candidate set $\mathcal{S}_{\text{nms}}$ using a physical radius r_{nms} . However, in low-overlap scenarios, true inliers shatter into topologically isolated islands within the compatibility graph $\mathcal{G}$. Because spatial NMS ignores graph topology, it often concentrates seeds in a single dominant cluster, neglecting valid fragmented inliers.

Graph Cluster Diversity via Topological Penalty. To explicitly bridge this fragmentation, we introduce an iterative topological penalty that forces sampling across disconnected components. Starting with $\mathcal{S}_0 = \emptyset$, at iteration t , we dynamically discount the score of each candidate $c_k \in \mathcal{S}_{\text{nms}}$:

$$\text{score}_k^{(t)} = \hat{s}_k - \lambda \cdot \max_{c_j \in \mathcal{S}_{t-1}} \exp\left(-\frac{d_{\mathcal{G}}(c_k, c_j)^2}{2\sigma_{\mathcal{G}}^2}\right), \quad (4)$$

where $d_{\mathcal{G}}$ is the shortest path distance in $\mathcal{G}$ (Eq. (1)), and the penalty term is set to 0 when $\mathcal{S}_{t-1} = \emptyset$. This kernel heav-

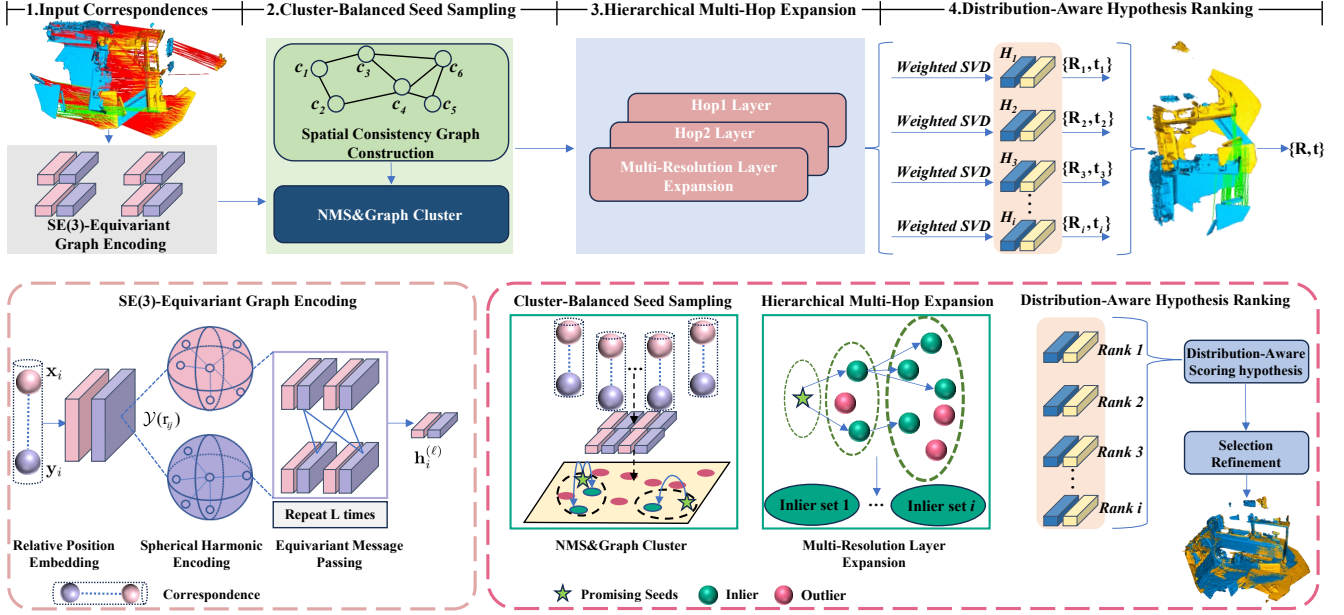


Figure 2. Method Overview. $SE(3)$ -equivariant graph encoding constructs a compatibility graph and learns equivariant embeddings to predict correspondence confidence scores $\{\hat{s}_i\}$. Cluster-balanced seed sampling leverages these scores to select M diverse seeds across fragmented graph clusters. Each seed then drives hierarchical multi-hop expansion, progressively discovering inliers through multi-resolution layers $\{\mathcal{L}_m^\ell\}_{\ell=0}^2$ via embedding retrieval, spectral filtering, and geometric verification. Finally, distribution-aware ranking evaluates spatial quality and alignment precision across all hypotheses to select the optimal transformation.

ily penalizes candidates topologically adjacent to already-selected seeds $\mathcal{S}_{t-1}$. We greedily select the highest-scoring candidate:

$$c_t = \arg \max_{c_k \in \mathcal{S}_{nms} \setminus \mathcal{S}_{t-1}} \text{score}_k^{(t)}, \quad \mathcal{S}_t = \mathcal{S}_{t-1} \cup \{c_t\}. \quad (5)$$

repeating until M seeds are collected to form the final seed set $\mathcal{S}_M$. Controlled by penalty strength λ and bandwidth σ_G , Eq. (4) dynamically depresses the sampling probability of an entire cluster once a seed is chosen. This guarantees that seeds comprehensively span diverse topological islands, providing a globally representative foundation for the subsequent multi-hop search (Sec. 3.4).

3.4. Hierarchical Multi-Hop Expansion

In rigid registration under extreme low-overlap, recovering the complete set of inliers from isolated seeds is notoriously difficult. While the physical objects being registered are strictly rigid, severe occlusions and sparse sampling cause true inliers to shatter into fragmented, isolated sub-graphs within the compatibility graph $\mathcal{G}$. Traditional global spectral methods [7, 21] or naive 1-hop seed-and-search paradigms [4, 18, 37] struggle in these fragmented scenarios. Attempting to bridge these topological gaps globally or in a single massive step inevitably engulfs massive outliers, causing a catastrophic drop in the Signal-to-Noise Ratio.

To safely retrieve fragmented inliers while suppressing overwhelming outlier noise, we formulate the search as a progressive trust-region expansion. Unlike generic proximity searches, our hierarchical multi-hop mechanism systematically explores $\mathcal{G}$ from high-confidence seeds outward, decomposing discovery into multi-resolution layers: 0-hop (seed anchor), 1-hop (strong structural neighbors), and 2-hop (extended geometric neighborhood).

Multi-Resolution Layer Expansion. Given the diverse seed set $\mathcal{S}_M$ obtained in Sec. 3.3, we construct a hypothesis m for each seed $s_m \in \mathcal{S}_M$. The 0-hop base layer is strictly initialized with the seed: $\mathcal{L}_m^0 = \{s_m\}$. Higher layers $\ell \in \{1, 2\}$ expand iteratively along the edges $\mathcal{E}$ of $\mathcal{G}$:

$$\mathcal{C}_m^\ell = \left\{ c \in \mathcal{C} \setminus \bigcup_{h < \ell} \mathcal{L}_m^h \mid \exists p \in \mathcal{L}_m^{\ell-1}, (c, p) \in \mathcal{E} \right\}. \quad (6)$$

To avoid redundant evaluations, we maintain a global unassigned pool $\mathcal{U}_m$ of correspondences not yet added to any layer.

At the 1-hop layer ($\ell = 1$), we first retrieve the top- k_1 candidates from $\mathcal{U}_m$ based on feature cosine similarity, where $k_1 = \lceil \sqrt{N} \log N \rceil$ and $N = |\mathcal{C}|$. To rigorously filter this candidate set, we extract its induced subgraph and compute the dominant eigenvector via power iteration. The top- k_2 candidates ($k_2 = \lceil \log_2 N \rceil$) with the highest eigen-

vector scores are then subjected to strict geometric verification ($\Delta d < \epsilon_d$ and $\Delta\theta < \epsilon_\theta$). Only verified correspondences are incorporated into $\mathcal{L}_m^1$, acting as trusted anchors for the subsequent expansion. At the 2-hop layer ($\ell = 2$), since spectral signals naturally weaken over longer topological distances, we perform a localized search from $\mathcal{L}_m^1$ into the remaining $\mathcal{U}_m$, relying directly on geometric verification.

Confidence Propagation. To dynamically guide this trust-region expansion, confidence scores propagate from verified parents to candidates. For a candidate $c \in \mathcal{C}_m^\ell$ ($\ell > 0$), its propagated confidence is:

$$\tilde{s}_c^{(\ell)} = \alpha \hat{s}_c + (1 - \alpha) \max_{p \in \mathcal{P}_c^{(\ell-1)}} \tilde{s}_p^{(\ell-1)}, \quad (7)$$

where $\mathcal{P}_c^{(\ell-1)} = \{p \in \mathcal{L}_m^{\ell-1} \mid (c, p) \in \mathcal{E}\}$ denotes its verified parent nodes in the preceding layer, $\hat{s}_c$ is the intrinsic confidence from Eq. (3), and α balances intrinsic quality with structural support. For the seed layer, we directly assign $\tilde{s}_{s_m}^{(0)} = \hat{s}_{s_m}$. This formulation ensures that high-confidence regions safely boost reliable topological paths while actively suppressing isolated outliers.

Ultimately, the complete inlier set for hypothesis m aggregates all progressive layers: $\mathcal{I}_m = \mathcal{L}_m^0 \cup \mathcal{L}_m^1 \cup \mathcal{L}_m^2$. The rigid transformation $\mathcal{T}_m = (\mathbf{R}_m, \mathbf{t}_m)$ is computed from $\mathcal{I}_m$ via weighted SVD [10] utilizing $\tilde{s}_c^{(\ell)}$ as weights, yielding a robust initial hypothesis for the final ranking stage.

3.5. Distribution-Aware Hypothesis Ranking

Spatial Quality Metric. Traditional hypothesis-and-verification methods typically rank transformations based solely on the Inlier Count (IC). However, in partial-overlap scenarios, this paradigm exhibits a critical vulnerability. Dense clusters of false-positive matches, frequently induced by perceptual aliasing on repetitive local structures, can easily accumulate a disproportionately high IC [18]. This inherently tricks the algorithm into selecting a geometrically degenerate and incorrect alignment. To robustly break this degeneracy, we introduce a distribution-aware spatial quality metric that explicitly evaluates both the quantity and the spatial coverage of the hypotheses.

For each of the M candidate hypotheses $\mathcal{T}_m = (\mathbf{R}_m, \mathbf{t}_m)$ generated in parallel from Sec. 3.4, we first identify its global valid inlier set evaluated over all input correspondences $\mathcal{C}$:

$$\mathcal{V}_m = \{c_i = (\mathbf{x}_i, \mathbf{y}_i) \in \mathcal{C} \mid \|\mathbf{R}_m \mathbf{x}_i + \mathbf{t}_m - \mathbf{y}_i\|_2 < \tau_d\}, \quad (8)$$

where τ_d is the strict inlier distance threshold. We then compute the comprehensive quality score:

$$Q(\mathcal{T}_m) = \text{IC}(\mathcal{T}_m) \cdot \left(1 - \exp\left(-\frac{\sigma_{\text{dist}}^m}{\tau_\sigma}\right)\right) \cdot \left(1 + \frac{\tau_d}{\bar{d}_m + \epsilon}\right), \quad (9)$$

where $\text{IC}(\mathcal{T}_m) = |\mathcal{V}_m|$ explicitly defines the global inlier count. Crucially, both the spatial spread σ_{dist}^m and the mean residual $\bar{d}_m$ are computed strictly on this verified global inlier set $\mathcal{V}_m$. Specifically, $\sigma_{\text{dist}}^m = \sqrt{\frac{1}{|\mathcal{V}_m|} \sum_{c_i \in \mathcal{V}_m} \|\mathbf{x}_i - \bar{\mathbf{x}}_m\|_2^2}$ measures the spatial dispersion of the source inliers around their centroid $\bar{\mathbf{x}}_m$, and $\bar{d}_m = \frac{1}{|\mathcal{V}_m|} \sum_{c_i \in \mathcal{V}_m} \|\mathbf{R}_m \mathbf{x}_i + \mathbf{t}_m - \mathbf{y}_i\|_2$. The exponential term, parameterized by τ_σ , heavily penalizes degenerate configurations with concentrated spatial distributions, while the precision term (with a small constant ϵ) rewards tighter local alignments.

Residual-Weighted Refinement. Even top-ranked hypotheses may suffer from minor outlier contamination near the boundary threshold τ_d . We select the top- K hypotheses based on $Q(\mathcal{T}_m)$ and refine each iteratively via weighted Procrustes [2, 10] operating on their respective valid sets $\mathcal{V}_m$:

$$\begin{aligned} \hat{\mathbf{R}}_m, \hat{\mathbf{t}}_m &= \arg \min_{\mathbf{R}, \mathbf{t}} \sum_{c_i \in \mathcal{V}_m} w_i^m \|\mathbf{R} \mathbf{x}_i + \mathbf{t} - \mathbf{y}_i\|_2^2, \\ w_i^m &= \exp\left(-\frac{\|\mathbf{R}_m \mathbf{x}_i + \mathbf{t}_m - \mathbf{y}_i\|_2^2}{2\tau_d^2}\right). \end{aligned} \quad (10)$$

The Gaussian weight w_i^m softly downweights correspondences with larger alignment residuals, preserving the dominance of highly accurate matches. Upon convergence, we re-evaluate the spatial quality $Q(\hat{\mathcal{T}}_m)$ for the refined hypotheses and select $\mathcal{T}^* = \arg \max_m Q(\hat{\mathcal{T}}_m)$ as the final registration result.

4. Experiment

Datasets. We conduct experiments on three benchmarks. 3DMatch [44] contains 1,623 indoor pairs from 8 RGBD sequences. 3DLoMatch [17] is a challenging subset with 1,781 low-overlap pairs at 10%-30% overlap. KITTI [14] contains 555 outdoor pairs from sequences 08-10.

Evaluation Criteria. Following DGR [10], we evaluate registration performance and correspondence quality using standard metrics. Registration is considered successful when $\text{RE} \leq 15^\circ$ and $\text{TE} \leq 30$ cm on 3DMatch and 3DLoMatch, and $\text{RE} \leq 5^\circ$ and $\text{TE} \leq 60$ cm on KITTI [4, 7]. Registration Recall measures the ratio of successful registrations. Rotation Error and Translation Error measure isotropic rotation error [15] and L_2 translation error, reported only on successful cases [4, 10]. Inlier Precision and Inlier Recall compute the ratio of correct correspondences to predicted and ground-truth correspondences, and F1 is their harmonic mean. A correspondence is correct when alignment error is below 10 cm for indoor and 60 cm for outdoor scenes [4]. For 3DLoMatch, we compute IP, IR, and F1 on all pairs regardless of registration success [4].

Implementation Details. Following standard protocols [4, 7], we downsample 3DMatch and 3DLoMatch with voxel

size $d_v = 5$ cm and extract FPFH [28] and FCGF [9] descriptors. For 3DLoMatch, we additionally employ Predator [17] and GeoTransformer [27] descriptors. For KITTI, we use $d_v = 30$ cm and extract FPFH and FCGF descriptors. Detailed computational complexity analysis is provided in the Appendix.

4.1. Evaluation on Indoor Scenes

We evaluate our method on 3DMatch [44] and 3DLoMatch [17] benchmarks using various feature descriptors and compare with both traditional and deep learning approaches.

Results on 3DMatch We compare our method against learning-based methods PointDSC [4], VBReg [18], DGR [10], and DHVR [20], as well as geometry-based approaches RANSAC [12], TEASER++ [36], SC²-PCR [7], MAC [45], FastMAC [46], Jiang *et al.* [19], and PCRegen [47]. Results in Tab. 1 show that using FPFH [28] descriptors, our method achieves 88.56% RR, surpassing PCRegen by 0.08%, with the best rotation error of 1.67°, which is 20.9% better than MAC. With FCGF [9] descriptors, our method matches PCRegen’s 93.96% RR while obtaining superior rotation error of 1.70° compared to 1.75°. Notably, our method requires only 0.10s runtime, identical to PointDSC and FastMAC, while achieving 11.0% higher RR and 23.4% lower RE than PointDSC with FPFH. This demonstrates that our multi-hop expansion effectively balances accuracy and efficiency across different feature qualities.

Table 1. Comparison results on 3DMatch [44] using FPFH [28] and FCGF [9] descriptors. **Bold**: best overall results. Underline: second-best among baseline methods.

Method	FPFH			FCGF			Time(s)
	RR↑	RE↓	TE↓	RR↑	RE↓	TE↓	
<i>Traditional Methods</i>							
FGR	40.91	4.96	10.25	78.93	2.90	8.41	0.89
RANSAC	66.10	3.95	11.03	91.44	2.69	8.38	2.86
TEASER++	75.48	2.48	7.31	85.71	2.73	8.66	0.03
SC ² -PCR	83.90	2.12	6.69	93.16	2.06	6.53	0.12
MAC	83.90	2.11	6.80	<u>93.72</u>	2.07	6.52	5.54
FastMAC	82.87	2.15	6.73	92.67	2.00	6.47	0.11
Jiang <i>et al.</i>	83.92	2.12	6.64	93.28	2.04	6.48	0.36
PCRegen	<u>88.48</u>	<u>1.70</u>	5.94	93.96	<u>1.75</u>	6.14	-
<i>Deep Learning Methods</i>							
DGR	32.84	2.45	7.53	88.85	2.28	7.02	1.53
DHVR	67.10	2.78	7.84	91.93	2.28	7.02	1.53
PointDSC	77.57	2.18	6.45	92.85	2.08	6.51	<u>0.10</u>
VBReg	82.57	2.14	6.77	93.53	2.04	6.49	1.01
MHopReg(<i>ours</i>)	88.56	1.67	<u>6.24</u>	93.96	1.70	<u>6.16</u>	<u>0.10</u>

Robustness to Different Features on 3DLoMatch. To evaluate robustness across different feature qualities, we conduct experiments on the challenging 3DLoMatch dataset using three descriptors: FCGF [9], Predator [17],

and GeoTransformer [27]. Results in Tab. 2 show consistent improvements across all descriptors. With FCGF, our method achieves 61.59% RR, outperforming VBReg [18] by 3.29% and MAC [45] by 1.74%, while obtaining the best IP of 48.12%, IR of 57.02%, and F1 of 52.19%. With Predator, our method achieves 72.19% RR, surpassing PCRegen [47] by 0.15%, with the lowest rotation error of 3.15° and highest IP of 64.02% and F1 of 64.16%. With GeoTransformer, our method achieves 79.48% RR, which is 0.57% over MAC, and the best rotation error of 2.66°, improving upon VBReg by 3.97%, with highest IP of 65.18% and F1 of 70.52%. These results demonstrate that our multi-hop expansion consistently discovers high-precision inliers across weak handcrafted features like FCGF and strong learning-based features like Predator and GeoTransformer.

Fig. 3 visualizes the progressive refinement on a challenging 3DLoMatch pair with 19.7% overlap. Starting from the seed hypothesis with 3.61% inlier ratio, our method progressively expands through Layer 1 to 36.3%, and Layer 2 to 98.32%, reducing outliers from 2722 to 6. This 103-fold improvement demonstrates how our multi-hop expansion effectively propagates geometric consistency through the correspondence graph, achieving sub-centimeter accuracy with RE of 0.01m and TE of 0.003m, even under extreme low-overlap conditions.

Qualitative results. Fig. 4 visualizes seed distribution learned by different strategies, which exhibits significant coverage differences on fragmented inlier clusters. Vanilla NMS concentrates seeds in dominant regions, while our cluster-balanced approach strategically spreads seeds across diverse graph clusters. This demonstrates that our graph-based penalty mechanism effectively ensures comprehensive hypothesis coverage, which is critical for robust registration in low-overlap scenarios.

4.2. Evaluation on Outdoor Scenes

We further validate our method on the KITTI odometry benchmark [14], which presents distinct challenges including large-scale environments, sparse point distributions, and planar structures. Following standard protocols [7, 19], we evaluate on 555 point cloud pairs with both FPFH [28] and FCGF [9] descriptors.

Results on KITTI. We evaluate on the KITTI dataset [14] using FPFH [28] and FCGF [9] descriptors, with results shown in Tab. 3. With FPFH, our method achieves 99.82% RR, matching PCRegen [47] and outperforming SC²-PCR [7] by 0.18%, though with higher translation error of 8.70 cm vs. 5.36 cm. With FCGF, our method achieves the best results across all metrics: 98.74% RR, 0.54% over Jiang *et al.* [19], lowest rotation error of 0.30°, 6.25% better than DHVR [20], and best translation error of 17.73 cm, 8.3% better than MAC [45]. Notably, our method maintains 0.13 s runtime, significantly faster than SC²-PCR at 0.31 s

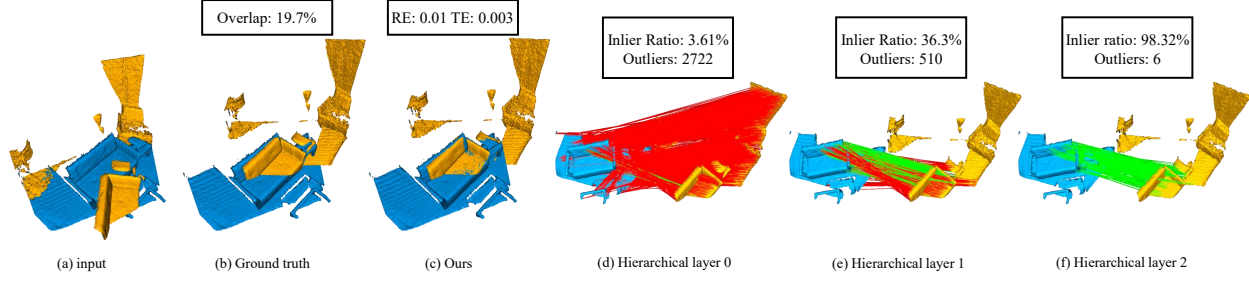


Figure 3. Progressive correspondence refinement through hierarchical multi-hop expansion.

Table 2. Quantitative results on 3DLoMatch Dataset.

	FCGF					
	RR↑	RE↓	TE↓	IP↑	IR↑	F1↑
FGR	19.99	5.28	12.98	27.63	19.16	19.98
RANSAC	46.38	5.00	13.11	40.70	44.61	42.02
SC ² -PCR	57.83	3.77	10.46	44.87	53.69	48.38
MAC	<u>59.85</u>	3.50	9.75	-	-	-
PCRegen	-	-	-	-	-	-
DHVR	54.41	4.14	12.56	41.96	38.60	39.22
DGR	43.80	4.17	10.82	42.22	38.96	39.05
PointDSC	56.09	3.87	10.39	44.51	52.38	47.57
VBReg	58.30	<u>3.32</u>	<u>10.08</u>	<u>47.24</u>	<u>56.05</u>	<u>51.26</u>
MHopReg(<i>ours</i>)	61.59	3.23	9.75	48.12	57.02	52.19
	Predator					
	RR↑	RE↓	TE↓	IP↑	IR↑	F1↑
FGR	35.99	4.77	11.64	47.18	38.76	39.10
RANSAC	64.85	4.28	11.04	56.44	65.68	60.01
SC ² -PCR	69.46	3.46	9.58	56.98	67.47	61.08
MAC	70.91	3.69	10.81	-	-	-
PCRegen	<u>72.04</u>	<u>3.17</u>	9.46	<u>63.95</u>	-	-
DHVR	65.41	4.97	12.33	54.75	54.66	53.70
DGR	59.46	3.19	10.01	51.38	54.24	51.62
PointDSC	68.89	3.43	9.60	56.55	67.52	60.82
VBReg	69.90	3.47	<u>9.56</u>	56.90	<u>67.84</u>	<u>61.89</u>
MHopReg(<i>ours</i>)	72.19	3.15	<u>9.56</u>	64.02	70.30	64.16
	GeoTransformer					
	RR↑	RE↓	TE↓	IP↑	IR↑	F1↑
FGR	-	-	-	-	-	-
RANSAC	77.48	3.37	9.69	<u>64.91</u>	73.98	68.68
SC ² -PCR	78.33	3.04	8.81	64.63	76.67	69.19
MAC	<u>78.91</u>	3.69	10.81	-	-	-
PCRegen	-	-	-	-	-	-
DHVR	73.83	4.49	10.21	61.06	71.85	64.21
DGR	-	-	-	-	-	-
PointDSC	77.82	3.00	8.71	63.65	76.87	68.39
VBReg	78.67	<u>2.77</u>	<u>8.62</u>	64.90	75.92	<u>69.97</u>
MHopReg(<i>ours</i>)	79.48	2.66	8.57	65.18	<u>76.81</u>	70.52

and PointDSC at 0.28 s while achieving superior accuracy, demonstrating efficient performance on large-scale outdoor scenes where point clouds contain 15K–20K points. This efficiency is directly attributed to our localized subgraph filtering, which safely avoids the exponential computational complexity of global matching. Furthermore, compared to deep learning baselines like DGR and VBReg, our approach operates over an order of magnitude faster while consis-

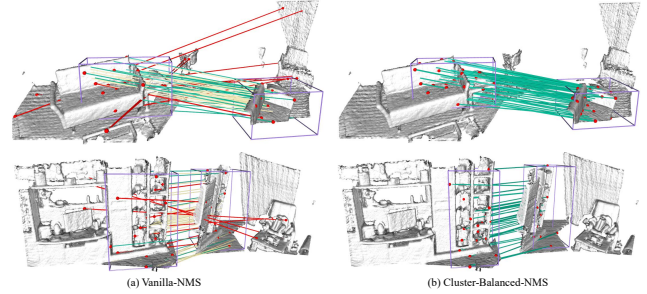


Figure 4. Visualizing seed selection strategies on two low-overlap 3DLoMatch pairs. The overlap regions are delineated with purple boxes. Selected seeds are marked with red spheres. Green/red lines indicate inlier/outlier correspondences discovered by seed-driven search, while light yellow lines show overlooked inliers within overlap zones. Note how vanilla NMS (a) concentrates seeds in dominant clusters, missing fragmented inlier regions, whereas our cluster-balanced approach (b) strategically distributes seeds across diverse graph clusters for comprehensive coverage.

tently delivering higher registration recall. Additionally, it substantially reduces the FCGF translation error from 20.69 cm in PointDSC to a highly competitive 17.73 cm, confirming its highly robust alignment capabilities.

Table 3. Results on KITTI dataset [14] using FPFH [28] and FCGF [9] descriptors.

Method	FPFH			FCGF			Time(s)
	RR↑	RE↓	TE↓	RR↑	RE↓	TE↓	
<i>Traditional Methods</i>							
FGR	5.23	0.86	43.84	89.54	0.46	25.72	3.88
RANSAC	74.41	1.55	30.20	80.36	0.73	26.79	5.43
TEASER++	91.17	1.03	17.98	95.51	0.33	22.38	0.03
SC ² -PCR	<u>99.64</u>	0.32	<u>7.23</u>	98.20	0.33	20.95	0.13
MAC	97.66	0.41	8.61	97.84	0.34	<u>19.34</u>	3.29
Jiang <i>et al.</i>	99.56	<u>0.34</u>	7.85	98.20	0.32	20.73	0.54
PCRegen	99.82	0.50	5.36	-	-	-	-
<i>Deep Learning Methods</i>							
DGR	77.12	1.64	33.10	96.90	0.34	21.70	2.29
DHVR	-	-	-	99.10	0.29	19.80	0.83
PointDSC	98.92	0.38	8.35	98.38	0.31	20.69	0.28
VBReg	98.92	0.45	8.41	98.02	0.32	20.91	1.34
MHopReg(<i>ours</i>)	99.82	0.39	8.70	<u>98.74</u>	<u>0.30</u>	17.73	<u>0.12</u>

4.3. Ablation Study

We conduct ablation experiments on 3DMatch and 3DLoMatch benchmarks using FPFH [28] descriptors to validate each component of our pipeline.

SE(3)-Equivariant Encoding. We verify the effectiveness of SE(3)-equivariant encoding by comparing with original learned confidence. Rows 1-2 show our encoding yields 0.24% higher RR on 3DMatch and 0.37% higher on 3DLoMatch, with significant IP gains of 8.15% and 0.58% respectively. This demonstrates that rotation-invariant embeddings provide crucial geometric consistency beyond feature similarity, particularly in low-overlap scenarios.

Seed Selection Strategy. We compare our Cluster-Balanced Seeding with NMS [4] and CSS [18]. Tab. 4 in rows 3-5 shows our method achieves 93.96% RR on 3DMatch and 61.59% on 3DLoMatch, outperforming NMS by 0.74% and 4.55%, and CSS by 0.61% and 4.21% respectively. The superior IP of 87.21% compared to 79.12% and 79.08% demonstrates that cluster-balanced seeding more effectively identifies high-quality seeds across diverse geometric regions.

Multi-Hop Expansion. We replace our Multi-Hop Expansion with Vote [18] and Two-Stage Selection (TS) [7]. Rows 6-8 show our method outperforms Vote by 0.68% and 4.31%, and TS by 0.55% and 2.63% in RR respectively. The improved IP of 87.21% compared to 82.74% and 78.97%, along with IR of 86.22% compared to 83.86% and 84.99% on 3DMatch, validate that our hierarchical expansion more effectively discovers inliers in fragmented clusters.

Hypothesis Ranking. We compare our Distribution-Aware Ranking with Inlier Count (IC) [12], TCD [8], and FS-TCD [8]. Rows 9-12 show our method achieves 1.11% and 5.44% RR improvement over IC respectively. While FS-TCD shows competitive RR of 93.72% and 60.70%, our method excels in IP on both benchmarks, achieving 87.21% compared to 79.05% on 3DMatch and 48.12% compared to 47.82% on 3DLoMatch, demonstrating that jointly evaluating inlier count, spatial spread, and geometric precision effectively prioritizes well-distributed transformations.

4.4. Efficiency-Accuracy Trade-off Analysis

To demonstrate the efficiency-accuracy trade-off of MHopReg, we conduct a scalability analysis on the KITTI dataset comparing against SC²-PCR [7] and PointDSC [4]. As shown in Fig. 5, we evaluate registration recall and processing time across varying input scales from 500 to 20K correspondences. At 500 correspondences, MHopReg achieves 97.82% recall, outperforming SC²-PCR with 54.05% by 43.77% and PointDSC with 97.47% by 0.35%. The substantial gap with SC²-PCR at limited scales highlights the advantage of our hierarchical expansion over global spectral methods. At 20K correspondences, MHopReg maintains 99.82% recall with 0.24s processing time,

Table 4. Ablation study on 3DMatch and 3DLoMatch benchmarks. Each block validates one core component of our method.

No.	Components	3DMatch			3DLoMatch		
		RR↑	IP↑	IR↑	RR↑	IP↑	IR↑
<i>SE(3)-Equivariant Encoding</i>							
1)	w/o Equivariance	93.72	79.06	86.34	61.22	47.54	56.79
2)	w/ Equivariance(ours)	93.96	87.21	86.22	61.59	48.12	57.02
<i>Seed Selection Strategy</i>							
3)	NMS	93.22	79.12	85.27	57.04	43.42	53.60
4)	CSS	93.35	79.08	85.34	57.38	43.47	53.80
5)	Cluster-Balanced(ours)	93.96	87.21	86.22	61.59	48.12	57.02
<i>Multi-Hop Expansion</i>							
6)	Vote	93.28	82.74	83.86	57.28	43.35	53.45
7)	Two-Stage	93.41	78.97	84.99	58.96	47.70	54.99
8)	Hierarchical Prop.(ours)	93.96	87.21	86.22	61.59	48.12	57.02
<i>Hypothesis Ranking</i>							
9)	IC	92.85	79.18	86.03	56.15	42.24	52.56
10)	TCD	93.35	79.26	85.02	58.14	45.87	54.59
11)	FS-TCD	93.72	79.05	85.95	60.70	47.82	57.10
12)	Dist.-Aware(ours)	93.96	87.21	86.22	61.59	48.12	57.02

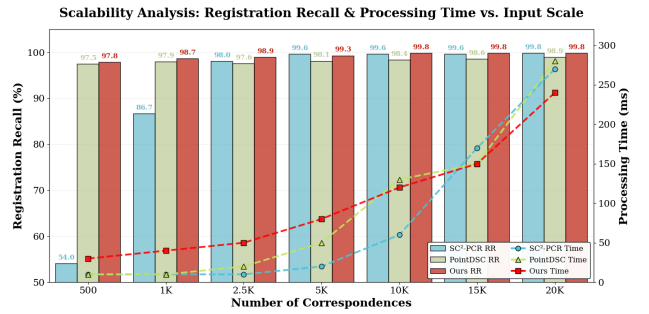


Figure 5. Scalability analysis on KITTI dataset with FPFH descriptors. The bar chart shows registration recall (left y-axis) and the line chart shows processing time (right y-axis) across different input scales. MHopReg maintains consistently high registration recall while demonstrating competitive efficiency and better scalability compared to baseline methods.

faster than SC²-PCR at 0.27s and PointDSC at 0.28s. The results demonstrate that MHopReg consistently maintains high accuracy across all scales while exhibiting superior scalability compared to both baselines.

5. Conclusion

We present MHopReg, a hierarchical multi-hop expansion framework for robust point cloud registration. Through cluster-balanced seeding and layer-wise propagation on the compatibility graph, our method efficiently discovers spatially coherent inliers. Distribution-aware ranking ensures optimal transformation selection by evaluating inlier quality and spatial coverage. Experiments demonstrate highly competitive performance with strong robustness to extreme outliers and low overlap. Future work will explore large-scale outdoor environments and cross-modality registration.

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